

How Green Buildings Pay Off: Covestro Case Study

► Covestro's eco-commercial building consumes over 70 per cent less energy than conventional buildings

Companies that invest in green practices for their factories or buildings derive significant environmental as well as financial profits. Energy-efficient polymer material manufacturer Covestro India's Eco-Commercial facility project at Greater Noida is a classic example of this. Green construction has enabled it to annually reduce over 70 per cent energy consumption and nearly 67 tonnes of greenhouse gas emission, while conserving natural resources. This case study shows how investing in green solutions led their LEED-Platinum rated facility to achieve these feats.

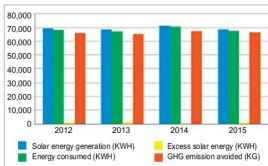
The setup

Covestro India underwent numerous transformations in terms of construction and infrastructure with major focus on optimising energy and resource consumption. Among the solutions embedded in the setup, the most significant were:

1. 56.7 kilowatt-peak grid-tied PV system with 210 watt-peak crystalline silicon solar modules and net-metering system
2. 75mm thick polyurethane foam (PUF) insulation for walls, roofs, slabs and ceilings
3. Roof rainwater storage
4. Wastewater treatment and

COST RECOVERY

Specifications	Details
Area of facility	8094sq.m (2-acre) land, 883sq.m (9500-sq.ft) super-built area
Total project cost	US\$ 4 million
Investment on eco-commercial technologies	₹ 11.7 million
Electricity savings by using solar grids	₹ 670,000 per year
Energy benefits due to PUF insulation	About 30 per cent
Expected ROI	11 years



Covestro's total energy production, excess energy stored, GHG reduction data during 2012-15

20KLD sewage treatment

Solar grids for renewable energy.

To generate electricity, Covestro installed PV systems with solar modules on its 445sq.m building roof, which were further connected to a distribution grid and a net-metering system. The average energy generation capacity of the grid is 88 megawatt-hour per year. Electricity generated by the solar system is distributed among the office and factory facilities by the grid. The excess power is stored in the grid and distributed throughout the facility during night or during shortage of solar power.

Insulation for energy-efficiency.

Covestro achieved energy-efficiency in its building with the help of PUF material. PUF is a poor conductor of heat, so any heat energy collected is not dissipated through PUF. Leveraging on this characteristic, Covestro structured the walls, facades, roofs and slabs with PUF. This cuts down the transfer of external heat to the indoors, keeping the office space cool. Reduced use of heavy-power cooling devices like air-conditioners

lowers the energy consumption.

Conserving water the smart way. Covestro minimises water wastage by installing low-flow fixtures throughout the office premises. There is an arrangement for storing rainwater on the roof, which is regularly used for landscape irrigation. Complete volume of wastewater is treated at

tertiary level for flushing and air-conditioning make-ups. Through these two steps, the enterprise cuts down potable water use. It also has a 20KLD sewage treatment plant based on sequential batch reactor technology.

The results

Electricity savings and reduced carbon footprint. The polyurethane foam restricts the use of air-conditioners by 30 per cent. The overall electricity savings using insulation coatings around the structures varies from 30 to 50 per cent.

According to the data shared by Covestro, their average yearly solar electricity generation is around 70,000 kilowatt-hour, of which up to 1400 kilowatt-hour is excess unconsumed energy.

Use of excess energy to power their facility at night instead of drawing power from the utility leads to major savings of non-renewable energy source. Moreover, this has reduced yearly greenhouse gas emission by almost 67,000kg.

Eventually, Covestro's eco-com-